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- as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))
- as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))

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(54) Title: SOLID STATE FORMS OF ZAVEGEPANT AND PROCESS FOR PREPARATION THEREOF

(57) Abstract: The present disclosure encompasses solid state forms of Zavegepant, in embodiments crystalline polymorphs of Zavegepant, processes for preparation thereof, and pharmaceutical compositions thereof.

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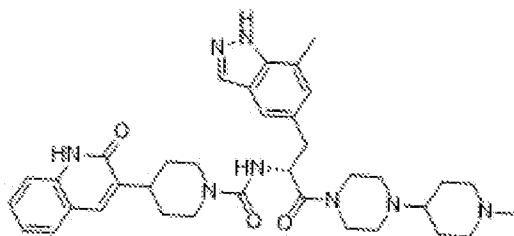
SOLID STATE FORMS OF ZAVEGEPANT AND PROCESS FOR PREPARATION THEREOF

FIELD OF THE DISCLOSURE

[0001] The present disclosure encompasses solid state forms of Zavegepant, in embodiments crystalline polymorphs of Zavegepant, processes for preparation thereof, and pharmaceutical compositions thereof.

BACKGROUND OF THE DISCLOSURE

[0002] Zavegepant, (R) N-(3-(7-methyl-1H-indazol-5-yl)-1-(4-(1-methylpiperidin-4-yl)piperazin-1-yl)-1-oxopropan-2-yl)-4-(2-oxo-1,2-dihydroquinolin-3-yl)piperidine-1-carboxamide, has the following chemical structure:



[0003] Zavegepant is a third generation, high affinity, selective and structurally unique, small molecule CGRP receptor antagonist, and it is developed for acute treatment of migraine or COVID-19 respiratory infection.

[0004] The compound is described in International Publication No. WO2011/123232.

[0005] Polymorphism, the occurrence of different crystalline forms, is a property of some molecules and molecular complexes. A single molecule may give rise to a variety of polymorphs having distinct crystal structures and physical properties like melting point, thermal behaviors (e.g., measured by thermogravimetric analysis (“TGA”), or differential scanning calorimetry (“DSC”)), X-ray diffraction (XRD) pattern, infrared absorption fingerprint, and solid state (^{13}C) NMR spectrum. One or more of these techniques may be used to distinguish different polymorphic forms of a compound.

[0006] Different salts and solid state forms (including solvated forms) of an active pharmaceutical ingredient may possess different properties. Such variations in the properties of different salts and solid state forms and solvates may provide a basis for improving formulation, for example, by facilitating better processing or handling characteristics, changing the dissolution profile in a favorable direction, or improving stability (polymorph as well as

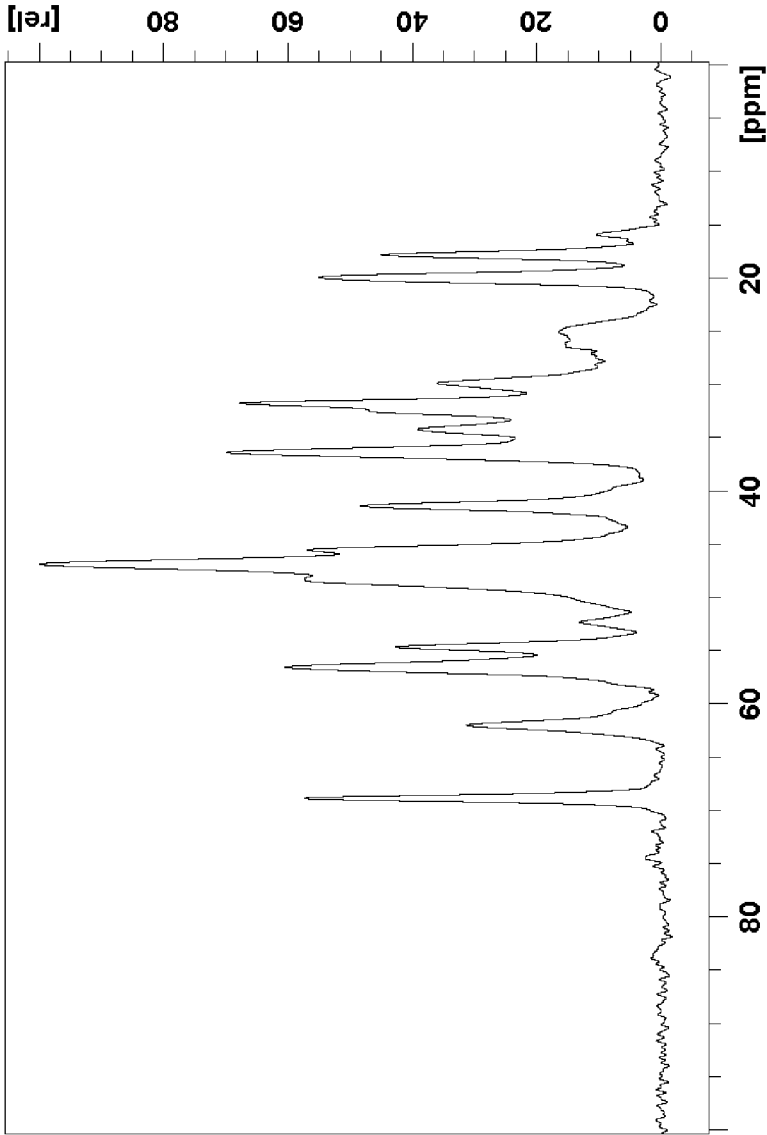


Figure 1c: ^{13}C solid state NMR spectrum of form A1 of Zavegepant (range from 100-0 ppm)

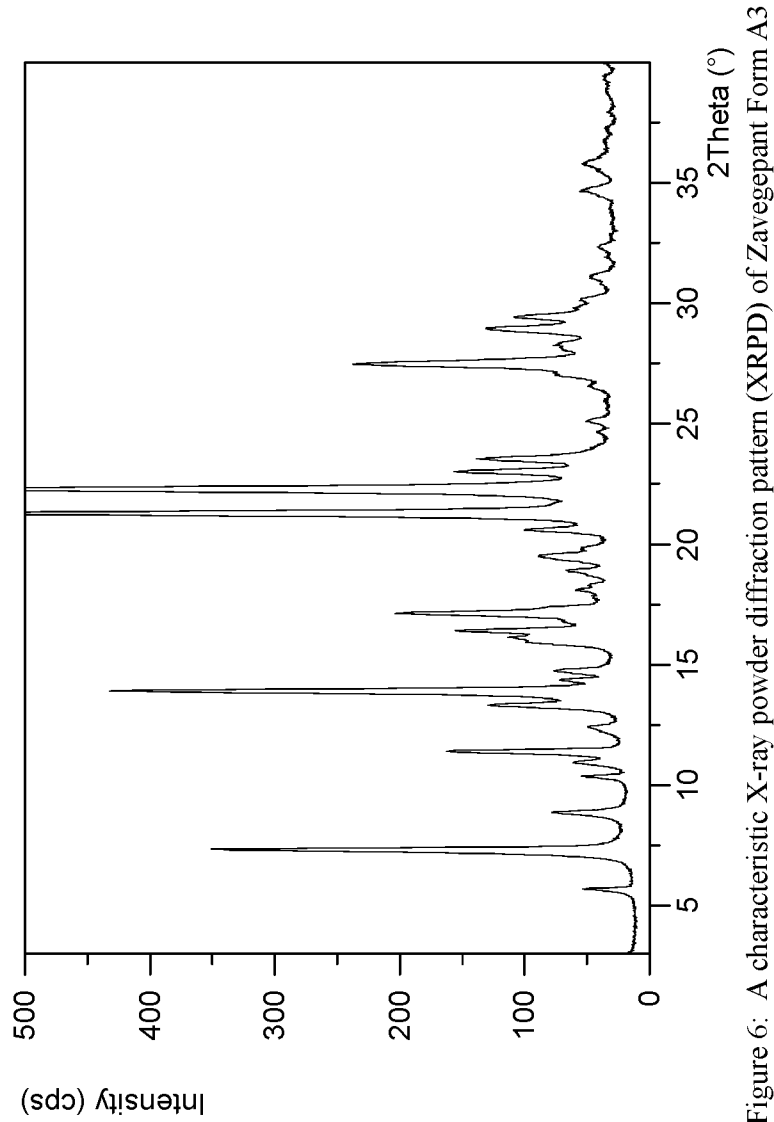


Figure 6: A characteristic X-ray powder diffraction pattern (XRPD) of Zavegepant Form A3

INTERNATIONAL SEARCH REPORT

International application No
PCT/US2022/023943

A. CLASSIFICATION OF SUBJECT MATTER

INV. C07D401/14 A61K31/4709 A61P31/12
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

C07D A61K A61P

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	R. O. CANN ET. AL.: "Selection of an Enantioselective Process for the Preparation of a CGRP Receptor Inhibitor.", ORGANIC PROCESS RESEARCH AND DEVELOPMENT, vol. 16, 14 November 2012 (2012-11-14), pages 1953-1966, XP055333905, DOI: 10.1021/op3003097 page 1959 column 2 and page 1965 column 2 -----	1-21
Y	WO 2011/123232 A1 (BRISTOL-MYERS SQUIBB COMPANY) 6 October 2011 (2011-10-06) cited in the application claims; examples -----	1-21



Further documents are listed in the continuation of Box C.



See patent family annex.

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"E" earlier application or patent but published on or after the international filing date

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"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

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